STPS Technologies

# British Manufacturing Company Becomes Industry Leader in Safety




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## Summary

Microchips are an important electronic component in the creation of devices such as the screen, phone, or computer that you are using to read this. These chips are made with even smaller, microscopic semiconductors, which are manufactured using highly specialized machines capable of the precision required to make these components.

SPTS Technologies, owned by the Keep Looking Ahead Foundation (KLA), designs, manufactures, sells, and supports capital equipment needed for companies to produce semiconductors and microelectronic devices. Located in Newport, Wales, UK, SPTS is an industry leader in safety and has received three ISO certifications in addition to recognition from numerous governing bodies. The International Organization for Standardization (ISO) recognized STPS for its excellence in its management systems for quality, environmental, and occupational health and safety. The capital equipment manufacturer has been actively reforming their organization to meet new standards and demonstrate their commitment to sustainability in their daily operations.

The company most closely aligns itself with the United Nations' sustainability development goals #3 and #13: good health and well-being, and climate action. The company has implemented a variety of initiatives to reduce their carbon footprint and promote employee wellness by creating an application its employees can use to more safely work with chemicals in the workplace as well as making various changes towards measuring and lowering their impact on the planet by reducing water and energy usage.

## Innovation

The internationally recognized standard for how companies manage their environmental impact, ISO 14001, was updated in 2015. Businesses that are serious about lowering their environmental impact follow this standard; its many rules ensure that a company is actively making changes to reduce its carbon footprint. SPTS is

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based in the United Kingdom, which has laws in place to make sure that businesses protect their employees under the Control of Substances Hazardous to Health (COSHH) regulations.

The changes made in regulations prompted SPTS to look into how their employees handle hazardous materials at work. Chris Webb, the environmental health & safety lead for SPTS, stated that “around about 2019, we did a review of the chemicals that we had on-site, including gases... We try to set time limits for having safety data sheets [that hold instructions for how chemicals should be handled] that are no more than 5 years old, but when we did the review, some of them were 35 years plus old.”

“SPTS is committed to the sustainable management of environment, health, and safety as a core business principle” (STPS.com, 2024). Maintaining the company's principles, Webb discovered that the safety information for the 500+ harmful materials they owned was disorganized and out of date. In 2019, Webb saw the need for change and initiated a project to create a company-wide platform for safety information. The program, as Webb describes, is an "internet-based system that serves as a register for all the chemicals that we use on-site...wherever you are in the world, all our employees can access the Internet, and they can access all the chemical data that they need."

By empowering its employees with the knowledge and skills on how to handle hazardous chemicals safely, Webb's initiative helps his company achieve the third sustainability development goal: good health and well-being.



## Inspiration

STPS has an extensive research and development team that is constantly innovating and creating innovative solutions for its customers that are also constantly innovating. This results in these teams periodically introducing new chemicals into the workplace to create machines capable of creating microelectronics. Because of the increasing number of chemicals, Webb recognized that the company needed a consolidated location where workers could easily access safety information for those chemicals, old and new.

Webb spoke on the implementation of their innovation, saying, “We have field support teams dotted across the world. If they need to work with a particular chemical, they can get all that information to them [using the application], and it's got all... the measures they need to put in place when they're using it.” These workers need to know what chemicals they are working with so as not to injure themselves and be able to properly handle these chemicals.

## Overall impact

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## Business benefit

SPTS was founded in 2009 with 40 years of manufacturing experience from companies such as Trikon Technologies, STS, and Aviza Technology. SPTS has committed itself to being a safe, sustainable, and industry-leading company, receiving numerous awards for their efforts over the years. Some of its numerous accolades include the RoSPA President's Award for Occupational Health and Safety (2024), the Queen's Award for Enterprise in Innovation (2020), and the SEMI Industry Leader Award for Diversity and Inclusion (2019).

As previously stated, SPTS was awarded for the health and safety measures they take to protect their employees. SPTS launched a hazardous substance program in 2019 over the course of approximately two years. They implemented occupational exposure monitoring, emissions monitoring, sensitive gas detection systems, and spill controls to ensure employee safety and measure the amount of pollutants the company is producing. These implementations are critical to employee retention, showing SPTS has made major advances in recent years to portray the importance they hold over their 590 employees, as the company benefits from employee loyalty. Their innovative approach to operational control serves to achieve sustainability development goals #3 and #13: good health and climate action, by illustrating their commitment to changing their operations in a manner that both helps their employees be safer and furthers their understanding of the impact they have on the planet.

SPTS engages with local stakeholders in environmental discussions to foster goodwill and improve community relationships, which are critical for long-term business success. By incorporating environmental strategies, SPTS has effectively positioned itself as a market leader, increasing competitiveness and driving innovation. Overall, their efforts have ensured regulatory compliance while also contributing to a more sustainable future, reinforcing their clearly stated commitment.

## Social and environmental benefit

SPTS Technologies has committed to improving society and the environment. SPTS has obtained ISO quality management certification, demonstrating their commitment to making a positive environmental impact and creating long-lasting change.

Although they are not required, businesses that believe they should maintain the highest standards choose to obtain ISO certifications. STPS has had this certification for a long time, but ISO was significantly revised in 2015, requiring companies to show a much greater interest in external stakeholders, meaning that SPTS includes local communities and governments in their environmental planning. SPTS must demonstrate continuous improvement each year in order to maintain its ISO certification, which the company has consistently achieved. Each year, senior management teams meet to agree on goals for the coming year, which leads to additional environmental and societal benefits.

In addition to the previously mentioned innovation, SPTS has made significant progress toward the thirteenth sustainability development goal, climate change, by installing energy-efficient water chillers and air filtering systems. These initiatives not only improve workplace safety, by reducing the chemicals used to filter the air, but also result in long-term energy cost savings, reflecting the company's values of business sustainability and environmental responsibility. SPTS has set the standard for

how businesses can positively contribute to the environment and society while still achieving business excellence!

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## Interview

Chris Webb, Environmental Health & Safety Lead

## Business information

### STPS Technologies

Newport, Wales, GB
**Business Website:** https://www.spts.com
**Year Founded:** 2009
**Number of Employees:** 501 to 1000

SPTS Technologies, a Keep Looking Ahead (KLA) company, specializes in the design, manufacturing, sales, and support of semiconductor manufacturing capital equipment. The company provides advanced wafer processing equipment and technologies to top semiconductor and microelectronics device manufacturers, as well as research institutions. It offers dedicated sales and customer support through a global network of offices and agents (SPTS, 2024).

Knowing "what" a business does is entirely different from "how" a business does it. The process of conducting an interview in this highly technical company was likely a first for all of us. Our interviewee, Chris Webb, unknowingly went through planning, operational, leadership, and control processes that were necessary to implement the innovation. Due to ISO and English regulatory changes, STPS did an internal analysis that prompted a change in how employees access outdated safety data. This is an example of how external and internal forces may lead to reform in a company. Webb showcases organizational change by describing how the mechanistic company created an application that empowers its employees to be able to access up-to-date safety data easily. Webb also stated how they measured the response to change using a compliance audit, showing how an organization may actively monitor their change in control. Meeting with the leader of innovation, we were able to see the passion he has for his work by his bright demeanor in the interview, reinforcing our ideas of what a leader may look and act like. There were surprising parts of the story, especially the 35-year-old data sheets; it shows the importance of doing routine audits on internal processes. Webb used a bit of jargon that we needed to put in more explainable terms, which led us to learn more about the types of chemicals that are common in the workplace at STPS as well as the type of work being done there. This led us to appreciate the changes the company is making for their employees and the environment, as hazardous chemicals can have a harmful impact on both groups. If ISO hadn't constricted their regulations, the company would have taken longer to learn the internal issues prompting the innovation we researched. We take away from our project that checks and balances, like regulation, are necessary to keep companies improving themselves, along with observing how the topics we learn can be applied in the real world.